Sprint Planning & MVP Communication

SG1 - The Green Mile
Sprint 3

Introduction

This is our group submission for the sprint planning and the MVP communication for Sprint 3.

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1. MVP Communication

After an intake with our client we came to a conclusion for a sprint goal and a MVP for this sprint:

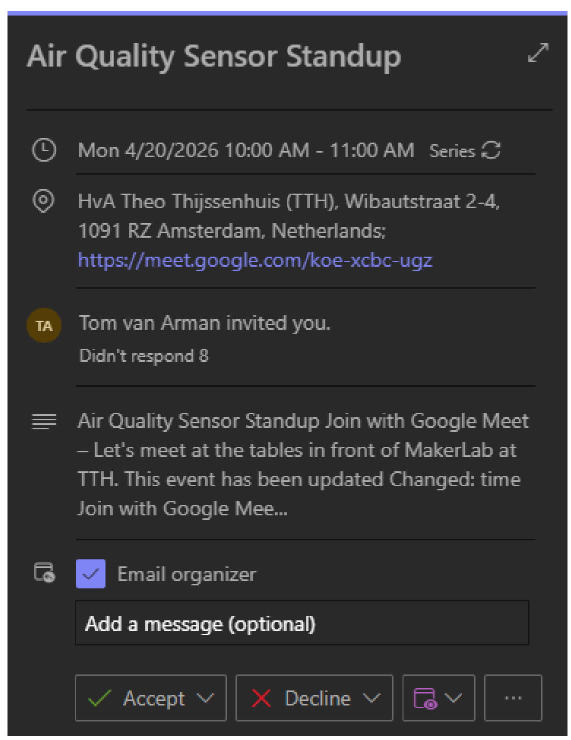
1. Embedded:
- Every box to communicate with MQTT
- PCB designed and ordered
- Prototype up and running for testing
- Assemble 4 prototype boxes
- Fixing sensor misreads and faulty communication

2. Back end:
- HTTP communication protocol replaced with MQTT to communicate with the back-end
- JSON payload examples and schema's for the cloud storage team
- Documentation for setting up own back-end and storage
- CI/CD

3. MVP:
- 4 prototype climate boxes (2 SIM card and 2 Wifi boxes)
- Each box communicates with MQTT
- Each box has the new internal design
- At least 2 boxes are ready for field tests

1.1 Meetings with client

We have a meeting with our client each Monday at 10 AM. This will be during the entire sprint. With our MVP meeting being flexible in the third week.



1.2 Feedback from client

PCB (Hardware Design)

- PCB is not required for the current prototypes
- Focus should remain on building and improving working prototypes
- PCB design can be developed in parallel as a separate task
- The Definition of Done includes a print-ready PCB design
- PCB development must not slow down or block prototype progress

Device Naming & Identity

- Replace the current name "CMB" with a clearer and more user-friendly name (e.g. *Airqual*)
- Apply the new naming consistently across all devices and documentation

User Stories Alignment

- Deep sleep improvements and sensor misread handling (assigned to Marnix)
- Ensuring correct and reliable data is sent to the backend (assigned to Hugo)
- Resolve overlap and conflict between these tasks to ensure aligned goals and implementation

Data Quality & Reliability

- Ensure the system sends accurate and trustworthy data
- Reduce incorrect readings and inconsistencies
- Improve stability of the data shown to users

Prototype Design Improvements

- Update the 3D-printed design to include a bracket for the new LED
- Ensure the design remains practical, durable, and easy to assemble

Overall MVP Focus

- Prioritize working, reliable prototypes over perfect hardware design
- Ensure clear structure, consistency, and usability across the system
- Keep development focused on delivering a functional and demonstrable MVP

2. Sprint Planning

The Sprint Backlog is ordered with User Stories (and Epics), highest priorities at the top. The team always selects the top User Story to work on next.

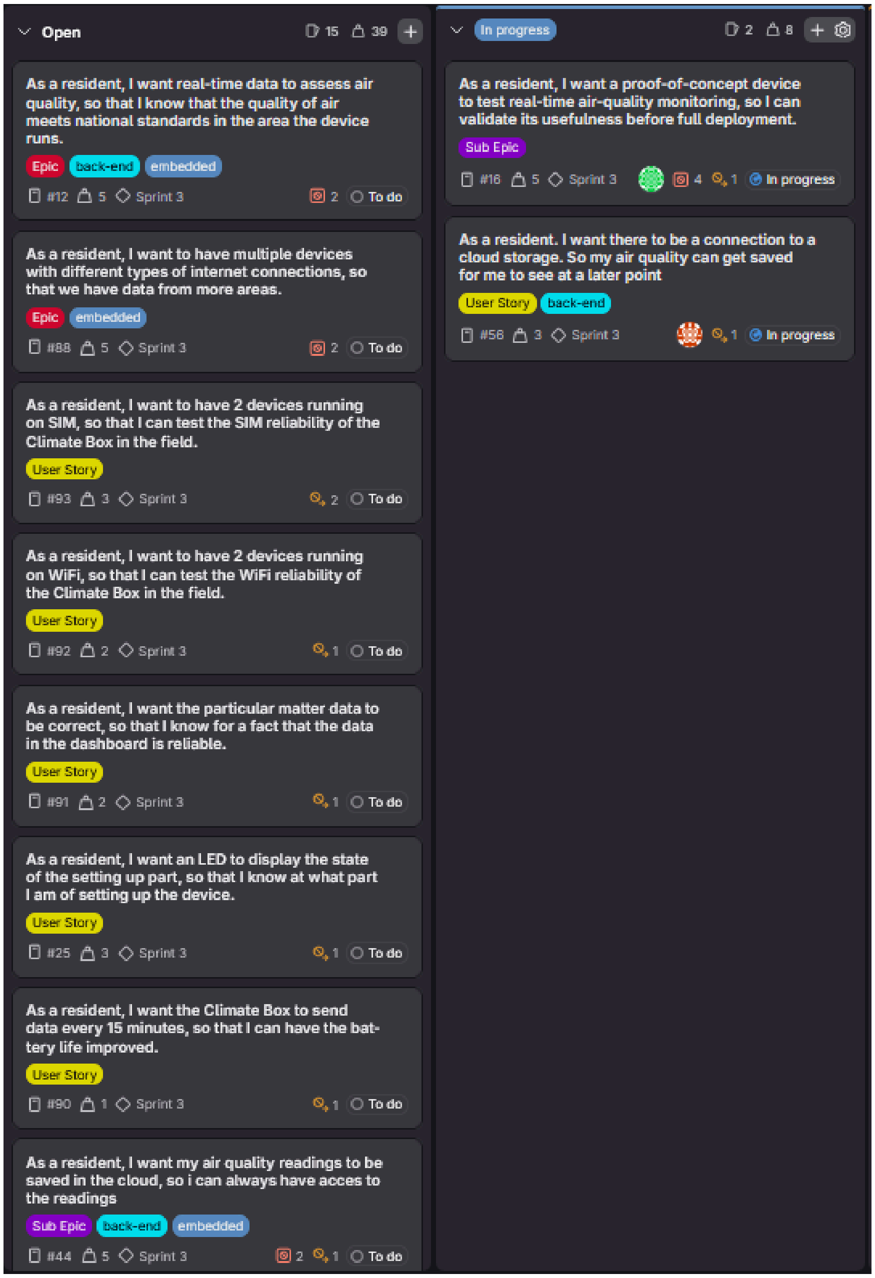
13-apr	11	Sprint 3
20-apr	12	Sprint 3
27-apr	-	Break
4-mei	13	Sprint 3

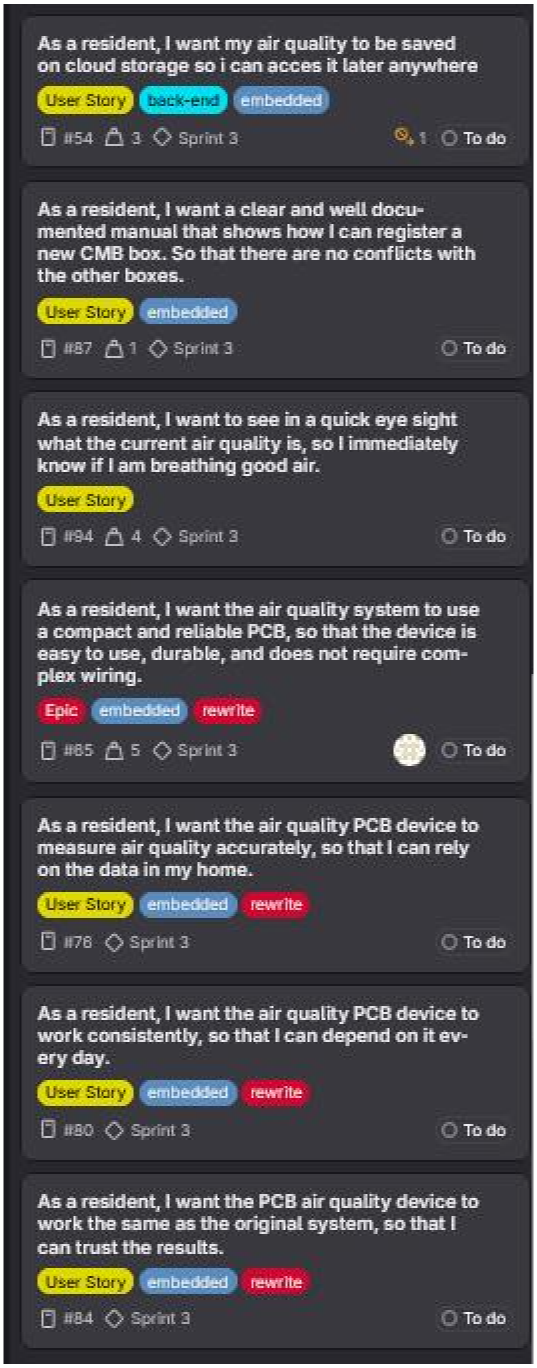
2.1 Sprint Backlog Requirements

- Weight
- Extensive description
- Acceptance criteria
- Definition of done
- Linked tasks

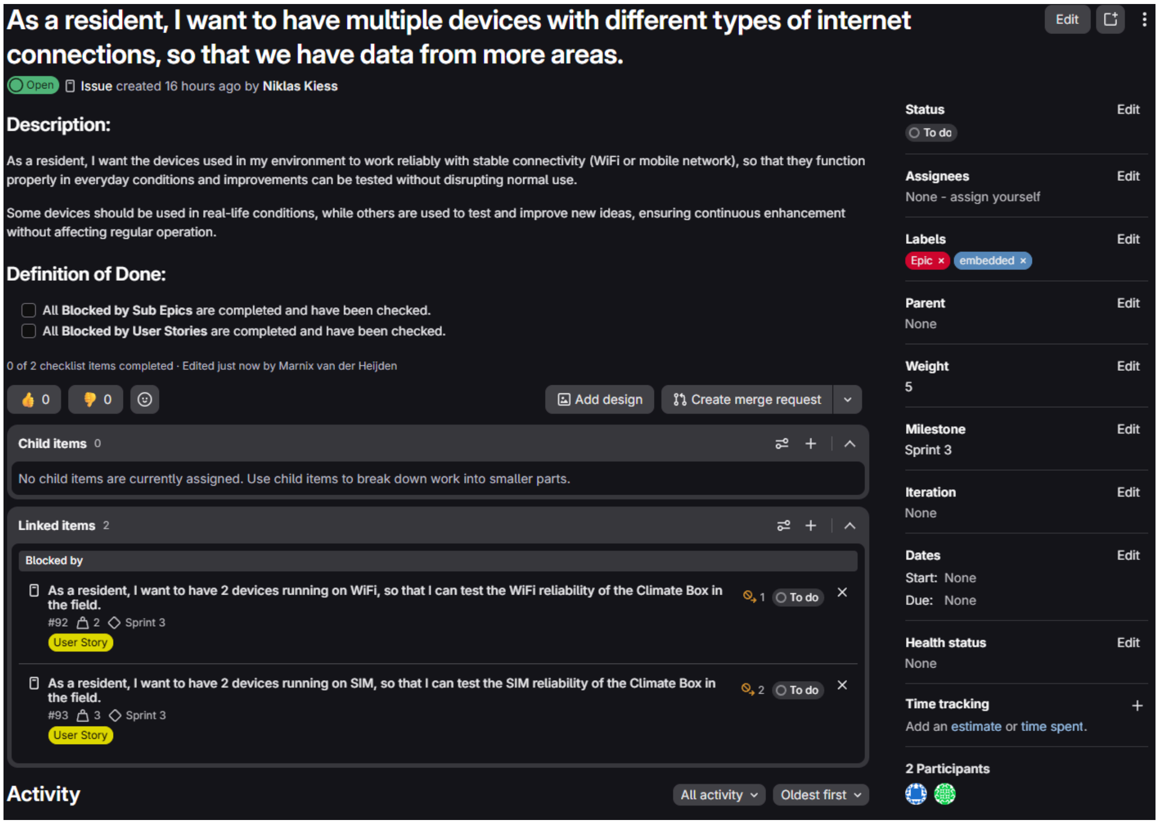
2.2 Sprint Backlog Evidence

- Screenshot of the Sprint Backlog board

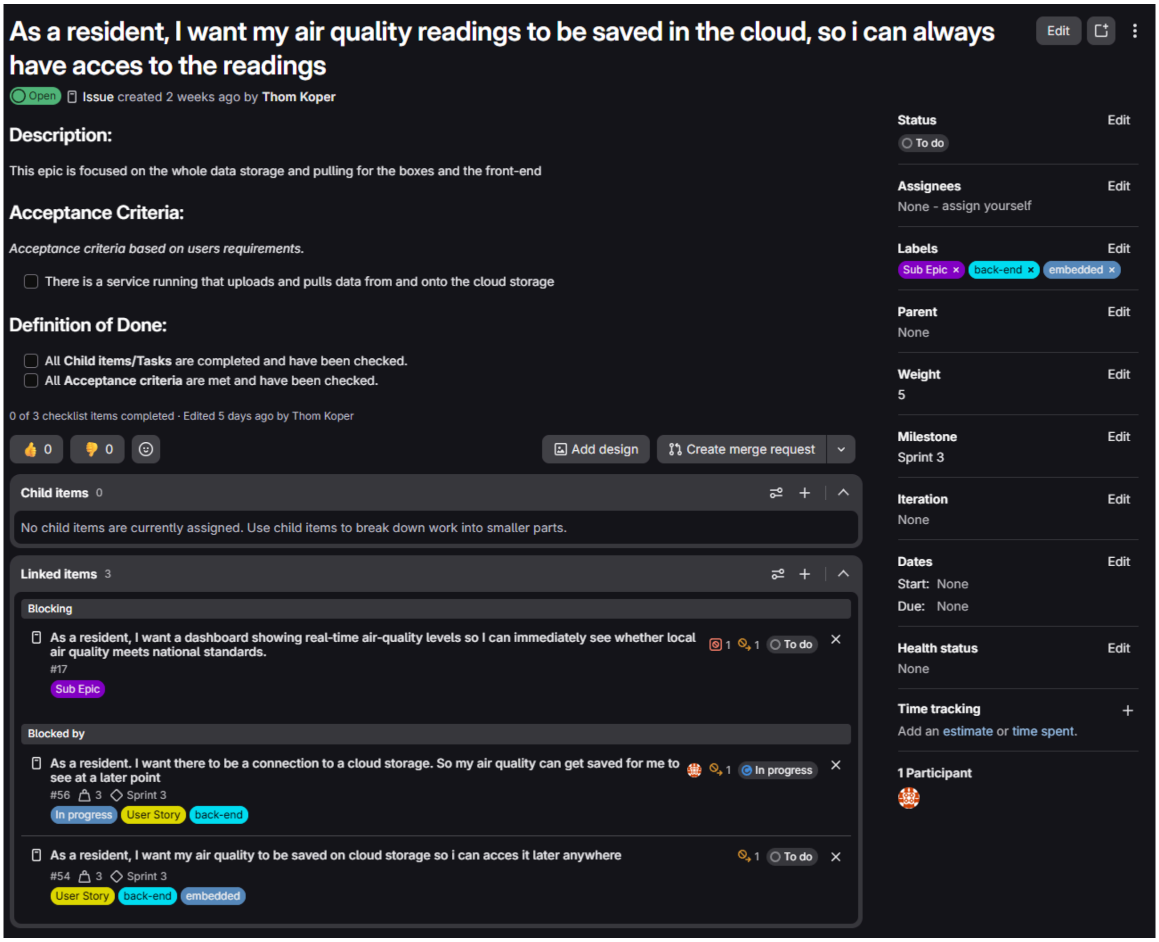



As a resident, I want my air quality to be saved on cloud storage so i can acces it later anywhere
User Story
back-end
embedded
#54 3 Sprint 3
1 To do
As a resident, I want a clear and well documented manual that shows how I can register a new CMB box. So that there are no conflicts with the other boxes.
User Story
embedded
#87 1 Sprint 3
To do
As a resident, I want to see in a quick eye sight what the current air quality is, so I immediately know if I am breathing good air.
User Story
#94 4 Sprint 3
To do
As a resident, I want the air quality system to use a compact and reliable PCB, so that the device is easy to use, durable, and does not require complex wiring.
Epic
embedded
rewrite
#85 5 Sprint 3
To do
As a resident, I want the air quality PCB device to measure air quality accurately, so that I can rely on the data in my home.
User Story
embedded
rewrite
#76 Sprint 3
To do
As a resident, I want the air quality PCB device to work consistently, so that I can depend on it every day.
User Story
embedded
rewrite
#80 Sprint 3
To do
As a resident, I want the PCB air quality device to work the same as the original system, so that I can trust the results.
User Story
embedded
rewrite
#84 Sprint 3
To do

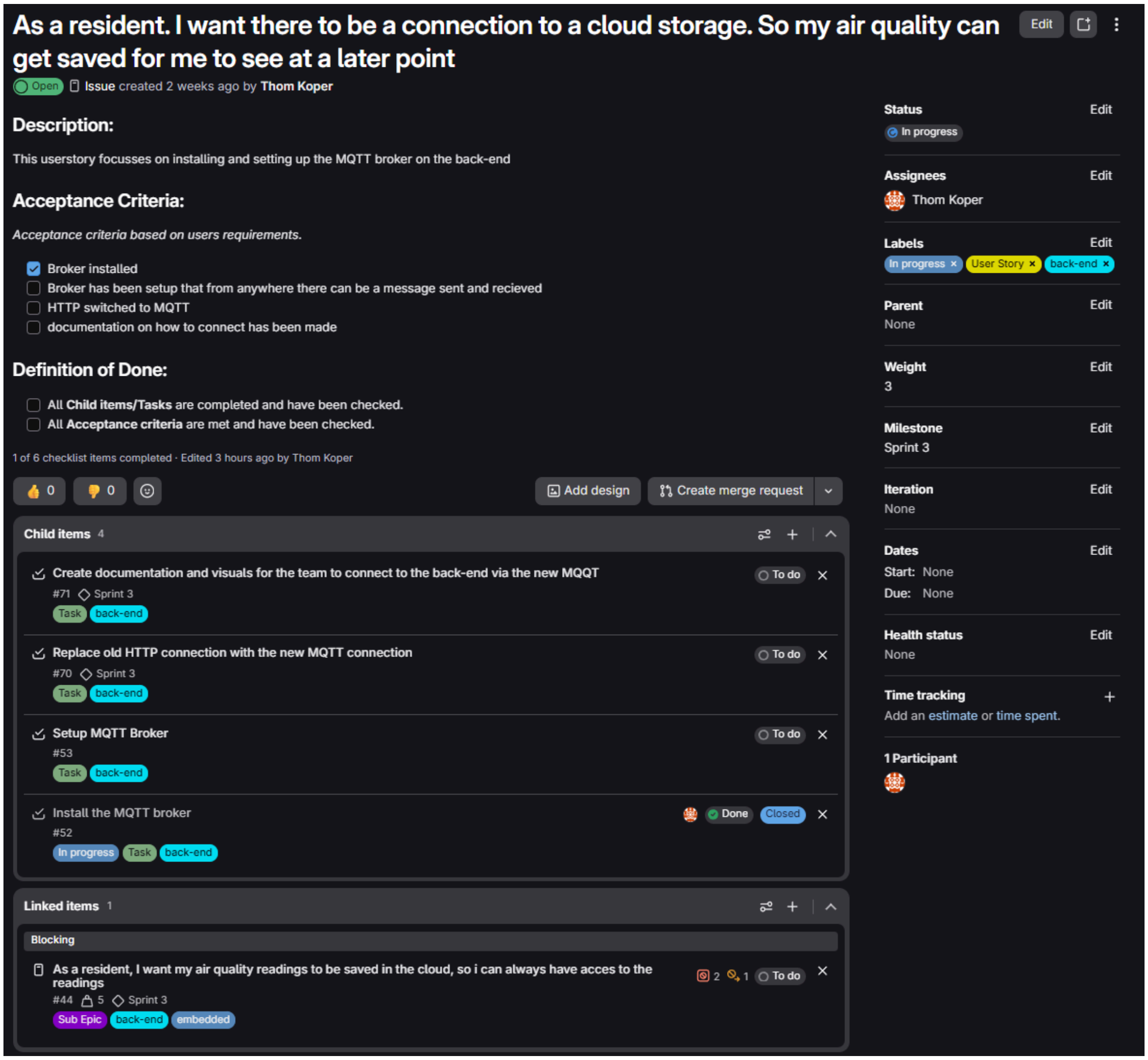
- Screenshot of a single Epic



- Screenshot of a single Sub Epic



- Screenshot of a single User Story



- Direct link to Sprint Backlog in Gitlab

https://gitlab.fdmci.hva.nl/studio/smart-cities/projecten/2025-2026-semester-2/group-project/sg1-the-green-mile/-/boards/41518?milestone_title=Sprint%203